1st Place: Aspect 6 – Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **Analysis**: This is the foundation for solving the problem. Students must accurately understand the mathematical relationship represented by the formula $D = \frac{dv}{60n}$ and know that when the target variable (v) is not the subject of the formula, algebraic rules need to be applied to transform it.
- **Manifestation in the Question**: Students need to recognize that to solve for v from $D = \frac{dv}{60n}$, they must perform a series of rule-based operations on algebraic equality: rearrange the formula to $v = \frac{60nD}{d}$. This requires them to master the algebraic transformation rule of "multiplying or dividing both sides of an equation by the same quantity" and understand the positions of variables d, v, n, D within the formula structure.

2nd Place: Aspect 7 – Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Analysis**: The core of this problem is handling **algebraic expressions** (formula transformation) and **numerical values** (substitution and calculation).
- **Manifestation in the Question**: The handling process includes two phases: 1) **Symbolic Manipulation**: Algebraically transform the formula $D = \frac{dv}{60n}$. 2) **Numerical Operation**: Substitute the given values $D = 50, n = 3, d = 25$ into the transformed formula $v = \frac{60 \times n \times D}{d}$ and perform multiplication and division to obtain the final result.

3rd Place: Aspect 8 – Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **Analysis**: A complete solution requires not only the final answer but also a clear demonstration of how the required formula is derived from the original one and the specific calculation steps.
- **Manifestation in the Question**: A student's solution should show the following intermediate steps:
  - Formula transformation: $D = \frac{dv}{60n} \Rightarrow dv = 60nD \Rightarrow v = \frac{60nD}{d}$ (or an equivalent process).
  - Numerical substitution: $v = \frac{60 \times 3 \times 50}{25}$.
  - Step-by-step calculation: e.g., first calculate the numerator $60 \times 3 = 180$, then $180 \times 50 = 9000$, finally $9000/25 = 360$.
  - Final answer: $v = 360$ mL.

4th Place: Aspect 12 – Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **Analysis**: Getting from reading the problem to the answer follows a clear, ordered multi-step procedure.
- **Manifestation in the Question**: A logical procedure is:
  1. **Identify**: Identify the known quantities (D, n, d) and the unknown quantity (v).
  2. **Model/Transform**: Solve for v from the given formula $D = \frac{dv}{60n}$ through algebraic manipulation.
  3. **Substitute**: Substitute the known numerical values into the new formula.
  4. **Calculate**: Perform the arithmetic operations.
  5. **Check & State**: Check the units (mL) and provide the final answer.

5th Place: Aspect 13 – Use an understanding of the context to guide or expedite the mathematical solving process

- **Analysis**: The medical context (infusion) and the units involved (drops/minute, drops/mL, hours, mL) provide the background for solving the problem. Understanding the physical meaning of these quantities helps verify the reasonableness of the answer.
- **Manifestation in the Question**: Students should understand that v is "volume" measured in mL. This reminds them to include the correct unit in the final answer. The context can also help with a rough check: Drip rate is 50 drops/minute for 3 hours (180 minutes), total drops = $50 \times 180 = 9000$ drops. If there are 25 drops per mL, then the volume should be $9000/25 = 360$ mL. This aligns with the algebraic solution, showing how the context can be used for quick verification.

6th Place: Aspect 1 – Perform a simple calculation

- **Analysis**: After substituting the values, a series of multiplication and division operations are required. These are basic but essential arithmetic calculations.
- **Manifestation in the Question**: Calculating $60 \times 3 \times 50/25$ requires accurately performing multiplication and division.

第1位：方面 6 - 理解并运用基于定义、规则和形式系统的构造

- **分析**：这是解题的基石。学生必须准确理解公式 $D = \frac{dv}{60n}$ 所代表的数学关系，并知道当求解目标（v）不是公式的主语时，需要应用代数规则对其进行变换。
- **题目体现**：学生需要认识到，为了从 $D = \frac{dv}{60n}$ 中解出 v，他们必须执行一系列基于代数等式的规则操作：将公式重排为 $v = \frac{60nD}{d}$。这要求他们掌握“方程两边同乘同除”的代数变换规则，并理解变量 d, v, n, D 在公式结构中的位置。

第2位：方面 7 - 处理数字、图形与统计信息、代数表达式和方程，以及几何表示

- **分析**：本题的核心是处理**代数表达式**（公式变换）和**数值**（代入计算）。
- **题目体现**：处理过程包括两个阶段：1) **符号操作**：将公式 $D = \frac{dv}{60n}$ 进行代数变换。2) **数值运算**：将给定的数值 $D = 50, n = 3, d = 25$ 代入变换后的公式 $v = \frac{60 \times n \times D}{d}$，并进行乘法、除法运算得出最终结果。

第3位：方面 8 - 阐明解决方案，展示和/或总结、呈现中间数学结果

- **分析**：一个完整的解答不仅需要最终答案，还应清晰地展示如何从原始公式推导出所需公式，以及具体的计算步骤。
- **题目体现**：学生的解答应展示以下中间步骤：
  - 公式变换：$D = \frac{dv}{60n} \Rightarrow dv = 60nD \Rightarrow v = \frac{60nD}{d}$（或等效过程）。
  - 数值代入：$v = \frac{60 \times 3 \times 50}{25}$。
  - 分步计算：例如，先计算分子 $60 \times 3 = 180$，再 $180 \times 50 = 9000$，最后 $9000/25 = 360$。
  - 最终答案：$v = 360$ mL。

第4位：方面 12 - 使用多步骤流程得出数学解答或结论

- **分析**：从读题到得出答案，遵循一个清晰、有序的多步流程。
- **题目体现**：一个逻辑流程是：
  1. **识别**：识别已知量 (D, n, d) 和未知量 (v)。
  2. **建模/变换**：从给定公式 $D = \frac{dv}{60n}$ 中，通过代数操作解出 v。
  3. **代入**：将已知数值代入新公式。
  4. **计算**：执行算术运算。
  5. **检查与表述**：检查单位（mL）并给出最终答案。

第5位：方面 13 - 利用对情境的理解来指导或加速求解过程

- **分析**：医疗情境（输液）和涉及的单位（滴/分钟、滴/毫升、小时、毫升）为解决问题提供了背景。理解这些量的物理意义有助于验证答案的合理性。
- **题目体现**：学生应理解 v 是“体积”，单位是 mL。这提醒他们在最终答案中必须包含正确的单位。情境也可能帮助进行粗略验证：滴速为 50 滴/分钟，持续 3 小时（180 分钟），总滴数为 $50 \times 180 = 9000$ 滴。如果每毫升 25 滴，那么体积应为 $9000/25 = 360$ 毫升。这与代数求解结果一致，展示了如何利用情境进行快速验证。

第6位：方面 1 - 执行简单计算

- **分析**：在代入数值后，需要进行一系列乘除运算。这些是基础但必要的算术计算。
- **题目体现**：计算 $60 \times 3 \times 50/25$ 需要准确执行乘法和除法运算。